Exploiting 2D MXene nanosheets for ultrasensitive plasmonic biosensing

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Abstract: We revealed the potential of exploiting 2D MXene nanosheets as plasmonic substrates for enhanced biosensing. The detection of BSA molecules based on MXene-on-Au substrate was successfully demonstrated with a detection limit down to 10 fM. © 2023 The Author(s)

1. Introduction

Possessing the merits of low cost, label free and real time detection capabilities , surface plasmon resonance (SPR) sensors have been widely used in studying molecule interactions and detecting biomarkers over these years [1,2]. However, the detection sensitivity of most traditional SPR sensors is not sufficient in cases where target analytes have low molecular weight or at a low concentration level [3]. To enhance the sensing signal, it is crucial to optimize the configuration of the plasmonic substrate. The thickness resolution of conventional SPR substrates based on Au or Ag materials are largely limited by the current fabrication technologies, which makes it difficult to tune the characteristics of the sensing interface and enhance the detection sensitivity. Here, we have proposed to add a layer of 2D MXene nanosheet on Au substrate to induce a steep phase variation and a giant lateral position shift for enhanced plasmonic sensing. We have demonstrated that our design can reach a high bulk sensitivity of up to 5.8×10^4 μm/RIU and realize the detection of BSA molecules with concentration level down to 10 fM. We envision that this 2D plasmonic substrate based on MXene nanosheets can serve as an effective tool in the ultrasensitive detection of various biomarkers.

2. Methods and materials

Here, we proposed a MXene-enhanced plasmonic sensing scheme based on phase induced lateral position shift measurement in Fig. 1. MXene is known to be a promising material for biosensing because of its excellent adsorption properties with a large specific surface area [4]. The MXene-on-Au metasurface was fabricated by spin coating a layer of 2D MXene nanosheet on top of a 40 nm gold thin film. Through simulation analysis using transfer matrix method (TMM), we have optimized the thickness of MXene layer to be 2.5 nm.

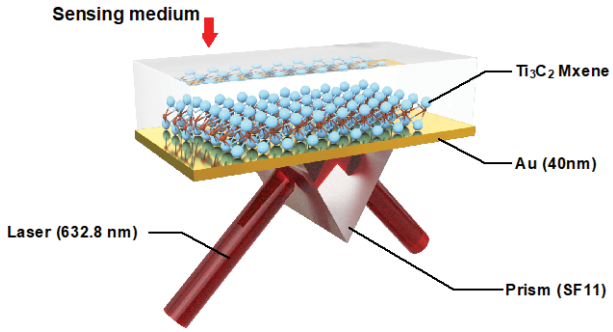

Fig.1 Schematic diagram of the plasmonic biosensing scheme based on MXene-on-Au substrate

The sensing scheme is based on the typical Kretschmann structure. In this system, a He-Ne laser (632.8 nm) was chosen to be the incident light source and divided into *p*-polarized and *s*-polarized light beams. A prism was fixed at the SPR angle to achieve minimum reflectivity and maximum lateral position shift change. We detected the lateral position shift signal through a high-resolution position sensor. The differential signal between *p*-polarized and *s*-polarized light was calculated to eliminate the environmental disturbance. A home-made microfluidic chamber was integrated with the plasmonic sensing substrate for reserving liquid samples to be detected.

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3. Results and discussion

According to simulation analysis, the optimized MXene-on-Au configuration will lead to a much deeper reflectivity dip in the angular spectrum. As shown in Fig. 2 (a), adding a layer of MXene will deepen the minimum reflectivity at the SPR angle from 0.08 to 9.3×10^{-6}, which in turn leads to a shaper change in phase signal. Moreover, Fig.2 (b) shows that the maximum GH shift, which is the higher order of phase signal ($L_{GH} = -\frac{1}{k}\frac{\partial\phi}{\partial\theta}$), will be further enhanced from 22 μm to 838 μm.

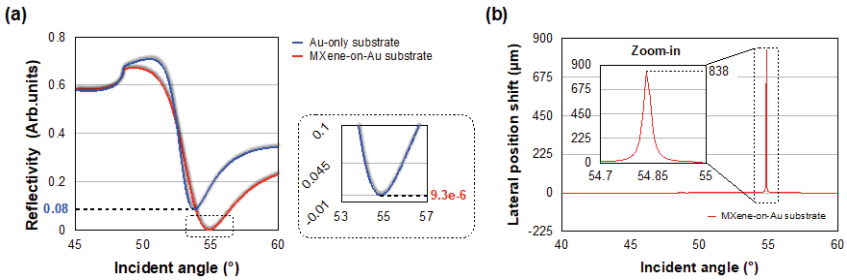

Fig.2 Simulation analysis based on MXene-on-Au substrate (a) Reflectivity spectra (b) Lateral position shift signal

As a standard calibration process, we have utilized this MXene-on-Au plasmonic metasurface to detect glycerol/water solution with different weight ratios and compared the results with traditional Au-only substrate. As shown in Fig.3 (a), the lateral position shift signal acquired based on MXene-on-Au substrate has a much higher signal response than Au-only substrate upon injection of glycerol solution with different concentration. The higher signal change indicates the higher bulk sensitivity (defined as $S = \frac{\Delta L_{GH}}{\Delta n}$) with respect to the refractive index change. In this case, the sensitivity can reach 5.8×10^4 μm/RIU for MXene-on-Au substrate.

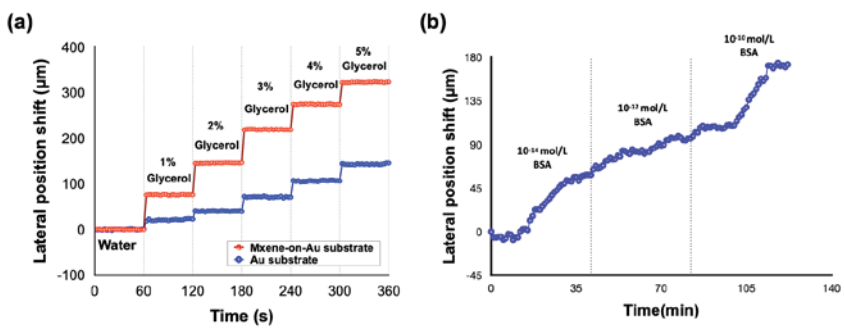

Fig.3 Experimental results (a) Detection of Glycerol solutions (b) Real-time detection of BSA solutions

In Fig. 3 (b), we have demonstrated the real-time and label-free detection of bovine serum albumin(BSA) molecules. As clearly depicted in the figure, the lateral position shift signal increased gradually and reached the stable state in 40 min upon the injection of BSA molecules, which indicated the successful detection of the binding process of BSA onto the substrate. More importantly, the application scenario based on our design can be further expanded to other target biomarkers, which is of great significance in the early-stage diagnostics of multiple diseases.

4. References

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